Do Now · retrieval

Boolean identities and De Morgan's laws

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Using h(k)=k MOD 5, what is the starting bucket for 17?

Q2

What does an adjacency list store for a vertex?

Q3

What is the result of TRUE AND FALSE?

Q4

Simplify A ∧ 1.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Negate the inputs — Distribute NOT across the brackets by negating A and B separately.

Core idea 2

Swap the operator — When applying De Morgan's law, AND changes to OR; OR changes to AND.

Core idea 3

Check equivalence — Both expressions produce the same output for all four combinations of A and B.

Key visual · complete concept

NOT A OR NOT B

A B	Result
00	**1**
01	**1**
10	**1**
11	**0**

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

Boolean identity — Rule for simplifying logical expressions.

De Morgan's law — Negate each operand and swap AND and OR.

complement — Logical NOT of a value or expression.

double negation — ¬¬A simplifies to A.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Simplify ¬(A ∨ B) and verify its truth table.

Follow the worked method

1. Predict and explain — Negate A and B separately and replace OR with AND.

2. Check and extend — Both expressions are true only when A=0 and B=0, proving equivalence.

Source model · final illustrated step

NOT A AND NOT B

A B	Result
00	**1**
01	**0**
10	**0**
11	**0**

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict and explain
De Morgan gives ¬A ∨ ¬(¬B).

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Simplify A ∨ 0.

Question 2

Use De Morgan to simplify ¬(P ∧ Q).

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Simplify ¬(A ∨ ¬B) and justify the final result.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

De Morgan's law allows us to remove a NOT from brackets without changing AND or OR.

Use key vocabulary from memory

1. Explain Boolean identity in your own words.

2. Explain De Morgan's law in your own words.

3. Explain complement in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.